

You can use <vector> from the STL to solve this problem, make sure to reference all the sources you used to solve this problem, which is from HackerRank who have several good technical interview challenges <https://www.hackerrank.com/challenges/array-left-rotation/problem>

Problem Outline

A *left rotation* operation on an array shifts each of the array's elements 1 unit to the left. For example, if 2 left rotations are performed on array [1,2,3,4,5], then the array would become [3,4,5,1,2]. Note that the lowest index item moves to the highest index in a rotation. This is called a **circular array**.

Given an array a of n integers and a number, d , perform d left rotations on the array. Return the updated array to be printed as a single line of space-separated integers.

Function Description

Complete the function *rotLeft*

rotLeft has the following parameter(s):

- $int\ a[n]$: the array to rotate
- $int\ d$: the number of rotations

Returns

- $int\ a'[n]$: the rotated array

Input Format

The first line contains two space-separated integers n and d , the size of a and the number of left rotations.

The second line contains n space-separated integers, each an $a[i]$.

Constraints

- $1 \leq n \leq 4,294,967,295$ // What is special about this number?
- $1 \leq d \leq n$
- $1 \leq a[i] \leq 4,294,967,295$

Sample Input

5 4

1 2 3 4 5

Sample Output

5 1 2 3 4

Explanation

When we perform $d=4$ left rotations, the array undergoes the following sequence of changes:

[12345] → [23451] → [34512] → [45123] → [51234]

Task 1 Code a solution to this problem in C++ (You don't need to upload your solution)

There is a solution available here <https://programmercave0.github.io/blog/2020/04/25/Left-Rotation-HackerRank-Challenge-Cpp-Implementation>

Discussion Task 2: For this task you will submit a short video (5 mins max) on Moodle explaining the solution given above and outlining how you would improve it.

Note: The max upload 50MB so as I cannot receive .zip or large attachments by email you may need to use a free video compression program to make your video smaller for Moodle upload. For example, "any video converter" <https://www.any-video-converter.com/products/video-converter-free/>